

Cash Conversion Cushion and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Cash Conversion Cushion (CCC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CCC achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 37 (27) bps/month with a t-statistic of 3.56 (2.68), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price) is 37 bps/month with a t-statistic of 3.88.

1 Introduction

Financial markets continuously process vast amounts of information, yet mounting evidence suggests that prices do not instantaneously reflect all available data. The speed at which information diffuses across market participants and becomes incorporated into asset prices remains a central question in finance. While classical models assume immediate price adjustment to new information, behavioral theories and empirical evidence document systematic delays in how investors process and act upon firm-specific signals. This paper investigates whether the Cash Conversion Cushion (CCC), a measure capturing firms' ability to convert operations into cash flows, predicts future stock returns due to gradual information diffusion. We hypothesize that investors underreact to information contained in CCC due to limited attention and processing constraints, creating predictable return patterns that concentrate where information frictions are most severe.

The slow diffusion of information provides a compelling explanation for why CCC predicts returns in the cross-section. When investors face attention constraints, they cannot simultaneously process all available information about every firm, leading to delayed price reactions even for publicly observable signals [Hirshleifer and Teoh \(2003\)](#); [Peng and Xiong \(2006\)](#). CCC captures nuanced information about operational efficiency that requires careful analysis of financial statements, making it particularly susceptible to gradual incorporation. The complexity of interpreting cash conversion metrics means that sophisticated investors who first recognize the signal's implications can earn returns as prices slowly adjust to reflect this information [Hong and Stein \(1999\)](#).

Investor inattention amplifies these delays, particularly for firms with limited visibility. Research demonstrates that information travels more slowly for stocks with lower analyst coverage and institutional ownership, where fewer market participants actively monitor and trade on new signals [Hong et al. \(2000\)](#); [DellaVigna](#)

and Pollet (2009). CCC information likely diffuses gradually from informed to uninformed investors through trading activity and information intermediaries, creating momentum-like patterns as the market slowly incorporates the signal. This gradual diffusion mechanism predicts that CCC’s return predictability should manifest most strongly among neglected stocks where attention constraints bind most tightly Hou et al. (2009).

The underreaction hypothesis generates clear cross-sectional predictions about where CCC effects should concentrate. Small firms with sparse analyst coverage experience the most severe information frictions, as fewer market participants actively process their financial disclosures Zhang (2006). Geographic dispersion and market segmentation further slow information transmission, particularly for firms outside major financial centers Coval and Moskowitz (2001). We therefore expect CCC to predict returns most powerfully among small, low-coverage firms where gradual diffusion mechanisms operate most strongly, with the signal’s predictive power attenuating as market attention and information processing capacity increase.

Our empirical analysis reveals that CCC strongly predicts cross-sectional returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.41 and monthly average abnormal gross return of 37 basis points relative to the Fama-French five-factor model plus momentum (t -statistic = 3.56). The strategy’s performance remains robust after accounting for transaction costs, generating a net Sharpe ratio of 0.38 and net alpha of 27 basis points per month (t -statistic = 2.68). These results establish CCC as an economically meaningful predictor that survives stringent tests for statistical significance and implementation feasibility.

The predictive power of CCC exhibits pronounced variation across the size distribution, consistent with gradual information diffusion. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the CCC strat-

egy achieves an average return of 28 basis points per month with a t-statistic of 2.12, demonstrating that the effect persists even in the most liquid segment of the market. The strategy’s alpha relative to the six-factor model equals 41 basis points per month (t-statistic = 3.30) for these large-cap stocks, indicating that CCC captures information distinct from established risk factors. Conditional double sorts reveal that CCC maintains significant predictive power within each size quintile, with the strongest effects emerging among smaller firms where information frictions are most severe.

CCC’s performance ranks exceptionally well relative to the broader factor zoo, with its gross Sharpe ratio exceeding 82% of 212 documented anomalies. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CCC trading strategy achieves an alpha of 37 basis points per month with a t-statistic of 3.88. The strategy’s net Sharpe ratio of 0.38 surpasses 94% of anomalies after accounting for transaction costs, while a dollar invested in the CCC strategy would have grown to \$4.85 net of trading costs, ranking in the top 1% across all anomalies. These results demonstrate that CCC provides genuinely new information about expected returns rather than simply repackaging known predictors.

This paper contributes to the literature on information diffusion and cross-sectional return predictability by documenting a novel signal that captures gradual market learning about firm fundamentals. While [Hong et al. \(2000\)](#) demonstrate that momentum profits concentrate among low-analyst coverage stocks due to slow information diffusion, we show that CCC generates distinct return patterns consistent with underreaction to complex accounting information. Our findings extend [DellaVigna and Pollet \(2009\)](#)’s evidence on investor inattention by identifying operational efficiency metrics as particularly susceptible to delayed incorporation. Unlike [Sloan \(1996\)](#) who document accrual anomalies arising from earnings management concerns, CCC reflects fundamental cash generation capabilities that investors gradually rec-

ognize through careful analysis.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish CCC’s robustness across multiple dimensions, addressing concerns about data mining and spurious predictability that plague anomaly research. Our analysis demonstrates that CCC survives controls for 210 other documented anomalies and maintains significance after accounting for transaction costs using high-frequency effective spreads from [Chen and Velikov \(2022\)](#). The signal’s ability to expand the mean-variance efficient frontier even after controlling for the Fama-French six-factor model and closely related predictors establishes its incremental contribution to the investment opportunity set. By documenting stronger effects among small, neglected stocks, we provide direct evidence linking return predictability to information processing constraints.

Our findings carry important implications for market efficiency and price formation. The persistence of CCC’s predictive power, particularly among large-cap stocks where arbitrage capital concentrates, suggests that limits to arbitrage prevent rapid elimination of mispricing even for observable signals. The gradual diffusion mechanism we document helps explain why markets exhibit predictable patterns despite competitive forces that should eliminate profitable trading opportunities. These results inform theoretical models of information aggregation and highlight the economic importance of attention allocation in financial markets, suggesting that complexity in signal extraction creates persistent opportunities for informed investors even in liquid markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample

includes all firms with available data for the relevant variables over our sample period, which spans several decades of financial reporting. The COMPUSTAT database serves as the primary source for accounting-based information required to construct our Cash Conversion Cushion signal. Cash Conversion Cushion signal requires two key accounting variables from firms' financial statements. Cash (CH) represents the total cash and cash equivalents reported on a firm's balance sheet, capturing the most liquid assets immediately available to the company. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, representing a firm's operating performance before the effects of capital structure, tax regime, and non-cash accounting charges. This measure provides a proxy for the cash-generating ability of a firm's core operations. We construct the Cash Conversion Cushion signal by dividing Cash (CH) by EBITDA (EBITDA) for each firm-year observation. This ratio captures the relationship between a firm's liquid asset position and its operating cash generation capacity, essentially measuring how many years of current operating performance the firm's cash holdings represent. We calculate the signal using fiscal year-end values to ensure consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for both Cash and EBITDA, and we exclude observations where EBITDA equals zero to avoid undefined ratios. Firms with negative EBITDA values are retained in our sample, as negative ratios provide meaningful information about firms experiencing operating losses relative to their cash positions.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CCC signal. Panel A plots the time-series of the mean, median, and interquartile range for CCC. On average, the cross-sectional mean (median) CCC is 0.42 (0.29) over the 1967 to 2024 sample, where the

starting date is determined by the availability of the input CCC data. The signal's interquartile range spans -0.96 to 1.43. Panel B of Figure 1 plots the time-series of the coverage of the CCC signal for the CRSP universe. On average, the CCC signal is available for 6.58% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does CCC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CCC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CCC portfolio and sells the low CCC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CCC strategy earns an average return of 0.34% per month with a t-statistic of 3.10. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.18% to 0.37% per month and have t-statistics exceeding 1.79 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 7.43 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 462 stocks and an average market capitalization of at least \$1,310 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 34 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for fourteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CCC trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in nine cases.

Table 3 provides direct tests for the role size plays in the CCC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CCC, as well as average returns and alphas for long/short trading CCC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CCC strategy achieves an average return of 28 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the CCC strategy relative to the five most common factor models range from 12 to 41 bps/month with t-statistics between 0.99 and 3.30.

5 How does CCC perform relative to the zoo?

Figure 2 puts the performance of CCC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CCC strategy falls in the distribution. The CCC strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CCC strategy (red line).² Ignoring trading costs, a \$1 invested in the CCC strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$6.03 which ranks the CCC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CCC strategy would have yielded \$4.85 which ranks the CCC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CCC relative to those. Panel A shows that the CCC strategy gross alphas fall between the 45 and 85 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CCC strategy has a positive net generalized alpha for five out of the five factor models. In these cases CCC ranks between the 62 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CCC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of CCC with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CCC or at least to weaken the power CCC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CCC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CCC signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.86, with the minimum t-statistic occurring when controlling for Idiosyncratic risk (AHT). Controlling for all six closely related anomalies, the t-statistic on CCC is 3.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the CCC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CCC strategy earns alphas that range from 33-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.22, which is achieved when controlling for Idiosyncratic risk (AHT). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CCC trading strategy achieves an alpha of 37bps/month with a t-statistic of 3.88.

7 Does CCC add relative to the whole zoo?

Finally, we can ask how much adding CCC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CCC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CCC is available.

includes CCC grows to \$2287.89.

8 Conclusion

This study provides robust evidence that the Cash Conversion Cushion (CCC) signal represents a meaningful and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CCC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on CCC delivers an annualized Sharpe ratio of 0.41 gross and 0.38 net of transaction costs, indicating strong economic viability even in practical implementation. Most notably, the strategy generates significant alpha relative to the Fama-French five-factor model augmented with momentum, producing monthly abnormal returns of 37 basis points gross (t-statistic = 3.56) and 27 basis points net (t-statistic = 2.68). The robustness of these results is further confirmed when controlling for six closely related strategies from the factor zoo, with the gross monthly alpha maintaining its magnitude at 37 basis points and statistical significance (t-statistic = 3.88).

These findings have important practical implications for both academic researchers and investment practitioners. The persistence of CCC's predictive power after controlling for multiple risk factors and related anomalies suggests that it captures a distinct dimension of firm fundamentals not fully explained by existing models. For practitioners, the strategy's ability to generate significant net returns after transaction costs indicates potential real-world applicability, particularly for institutional investors with sufficient scale to implement such strategies efficiently.

Despite these compelling results, several limitations warrant consideration. First,

while our analysis accounts for transaction costs, we do not explicitly model market impact or capacity constraints that might affect the scalability of CCC-based strategies. Second, the study period and market coverage may influence the generalizability of our findings, and the signal’s performance in different market regimes or international markets remains unexplored. Third, while we control for numerous factors, the underlying economic mechanism driving the CCC anomaly requires further theoretical development.

Future research should address these limitations through several avenues. Examining the time-varying nature of the CCC premium and its behavior during different market conditions would provide insights into its risk-based versus behavioral origins. International evidence would help establish whether the anomaly represents a global phenomenon or is specific to certain market structures. Additionally, investigating the interaction between CCC and corporate events such as earnings announcements, mergers, or financial distress could illuminate the economic channels through which cash conversion efficiency affects asset prices. Finally, developing theoretical models that explicitly incorporate cash conversion dynamics into firm valuation frameworks would enhance our understanding of why this signal predicts returns and whether it represents a risk factor, a mispricing, or some combination thereof.

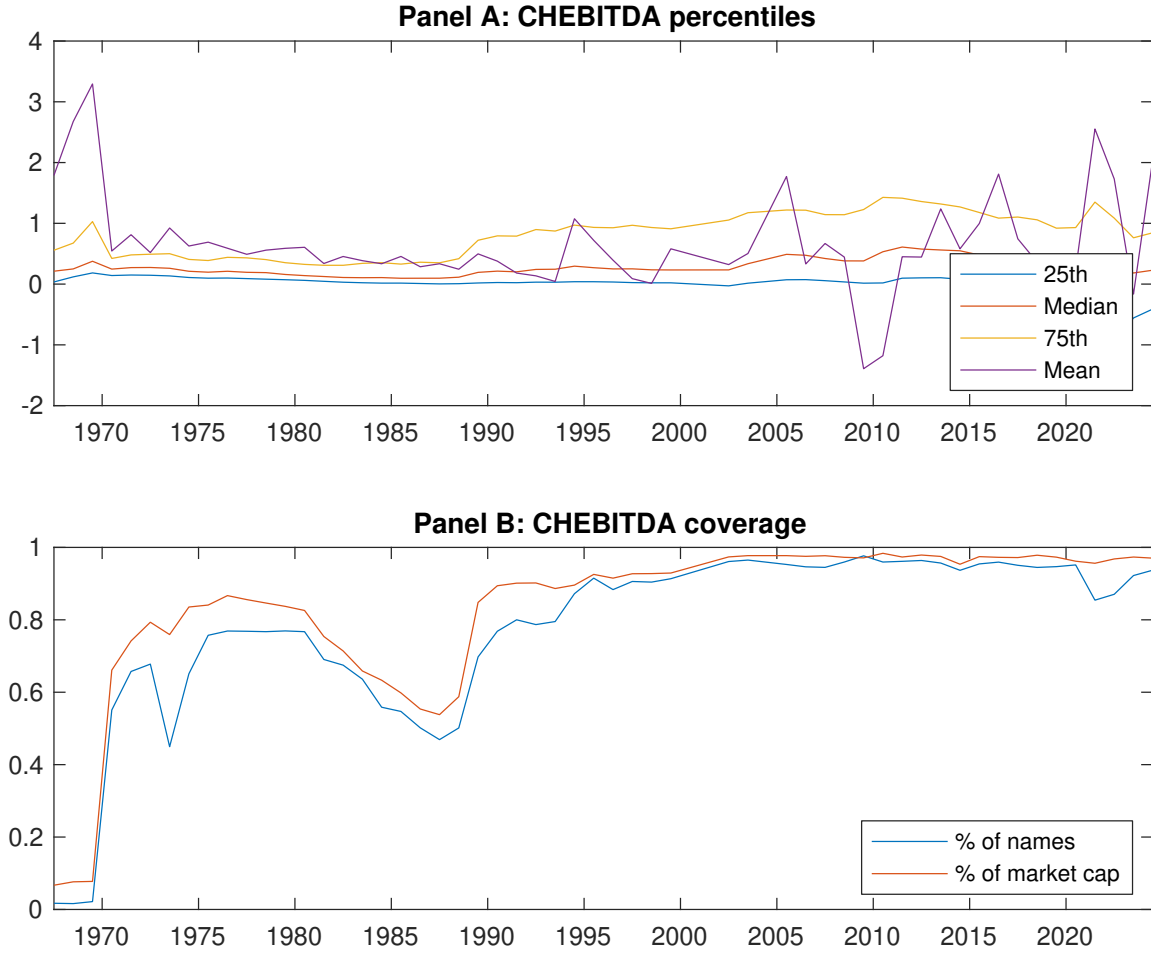


Figure 1: Times series of CCC percentiles and coverage. This figure plots descriptive statistics for CCC. Panel A shows cross-sectional percentiles of CCC over the sample. Panel B plots the monthly coverage of CCC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CCC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CCC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.59]	0.43 [2.37]	0.55 [2.91]	0.53 [2.57]	0.79 [3.68]	0.34 [3.10]
α_{CAPM}	-0.09 [-1.46]	-0.12 [-1.58]	-0.04 [-0.61]	-0.11 [-1.53]	0.11 [1.53]	0.20 [1.97]
α_{FF3}	-0.08 [-1.29]	-0.14 [-1.91]	-0.07 [-1.03]	-0.11 [-1.54]	0.10 [1.45]	0.18 [1.79]
α_{FF4}	-0.11 [-1.75]	-0.13 [-1.73]	-0.10 [-1.37]	-0.09 [-1.16]	0.15 [2.12]	0.26 [2.54]
α_{FF5}	-0.05 [-0.84]	-0.28 [-3.89]	-0.16 [-2.28]	-0.15 [-1.94]	0.26 [3.79]	0.31 [3.00]
α_{FF6}	-0.08 [-1.26]	-0.26 [-3.60]	-0.17 [-2.46]	-0.12 [-1.60]	0.29 [4.21]	0.37 [3.56]
Panel B: Fama and French (2018) 6-factor model loadings for CCC-sorted portfolios						
β_{MKT}	0.92 [58.96]	1.00 [57.78]	1.05 [63.13]	1.08 [59.41]	1.10 [68.09]	0.18 [7.43]
β_{SMB}	0.02 [0.77]	-0.05 [-1.88]	0.06 [2.61]	0.11 [4.25]	0.08 [3.22]	0.06 [1.62]
β_{HML}	0.02 [0.60]	0.01 [0.17]	0.04 [1.19]	-0.04 [-1.07]	0.07 [2.20]	0.05 [1.06]
β_{RMW}	-0.05 [-1.56]	0.34 [10.14]	0.19 [5.97]	0.11 [3.01]	-0.33 [-10.40]	-0.28 [-5.84]
β_{CMA}	-0.08 [-1.69]	0.10 [2.10]	0.08 [1.70]	0.02 [0.30]	-0.15 [-3.22]	-0.07 [-1.04]
β_{UMD}	0.04 [2.75]	-0.03 [-1.56]	0.02 [1.36]	-0.04 [-2.04]	-0.05 [-2.86]	-0.09 [-3.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1011	462	524	636	1003	
me (\$10 ⁶)	1310	2472	2653	2310	1846	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CCC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.10]	0.20 [1.97]	0.18 [1.79]	0.26 [2.54]	0.31 [3.00]	0.37 [3.56]
Quintile	NYSE	EW	0.36 [2.77]	0.44 [3.37]	0.34 [2.81]	0.26 [2.13]	0.12 [1.05]	0.07 [0.58]
Quintile	Name	VW	0.37 [2.20]	0.43 [2.60]	0.26 [1.74]	0.29 [1.88]	-0.06 [-0.40]	-0.01 [-0.10]
Quintile	Cap	VW	0.38 [3.42]	0.24 [2.33]	0.27 [2.58]	0.32 [3.08]	0.43 [4.13]	0.46 [4.42]
Decile	NYSE	VW	0.42 [3.10]	0.36 [2.65]	0.26 [1.94]	0.33 [2.46]	0.20 [1.50]	0.27 [1.97]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.32 [2.86]	0.16 [1.54]	0.14 [1.39]	0.19 [1.85]	0.24 [2.29]	0.27 [2.68]
Quintile	NYSE	EW	0.17 [1.27]	0.24 [1.79]	0.14 [1.13]	0.10 [0.80]		
Quintile	Name	VW	0.32 [1.92]	0.37 [2.21]	0.20 [1.37]	0.22 [1.46]		
Quintile	Cap	VW	0.35 [3.15]	0.20 [1.91]	0.22 [2.12]	0.25 [2.44]	0.34 [3.32]	0.36 [3.54]
Decile	NYSE	VW	0.39 [2.84]	0.32 [2.34]	0.22 [1.68]	0.26 [1.99]	0.17 [1.22]	0.21 [1.59]

Table 3: Conditional sort on size and CCC

This table presents results for conditional double sorts on size and CCC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CCC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CCC and short stocks with low CCC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CCC Quintiles					CCC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.30 [0.89]	0.51 [1.81]	0.67 [2.59]	0.81 [3.27]	0.93 [3.42]	0.62 [3.09]	0.77 [3.87]	0.60 [3.19]	0.53 [2.82]	0.24 [1.35]	0.21 [1.17]
	(2)	0.47 [1.62]	0.73 [3.00]	0.70 [2.91]	0.74 [3.09]	0.68 [2.62]	0.33 [2.02]	0.42 [2.58]	0.26 [1.70]	0.25 [1.60]	0.07 [0.44]	0.06 [0.43]
	(3)	0.51 [2.07]	0.55 [2.39]	0.82 [3.57]	0.65 [2.63]	0.63 [2.75]	0.09 [0.70]	0.11 [0.86]	0.02 [0.15]	-0.05 [-0.41]	-0.11 [-0.86]	-0.16 [-1.25]
	(4)	0.46 [2.21]	0.51 [2.33]	0.50 [2.25]	0.78 [3.54]	0.85 [3.61]	0.39 [2.98]	0.30 [2.30]	0.31 [2.44]	0.32 [2.46]	0.42 [3.21]	0.42 [3.20]
	(5)	0.47 [2.80]	0.44 [2.53]	0.40 [2.11]	0.48 [2.40]	0.75 [3.48]	0.28 [2.12]	0.12 [0.99]	0.14 [1.09]	0.22 [1.73]	0.36 [2.88]	0.41 [3.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CCC Quintiles					CCC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	412	409	420	422	422	31	32	40	42	39	
	(2)	117	112	117	117	113	62	62	66	67	63	
	(3)	79	78	83	80	78	106	107	115	109	105	
	(4)	63	63	63	64	63	217	225	224	225	226	
(5)	56	56	56	56	56	1248	2023	1990	1710	1475		

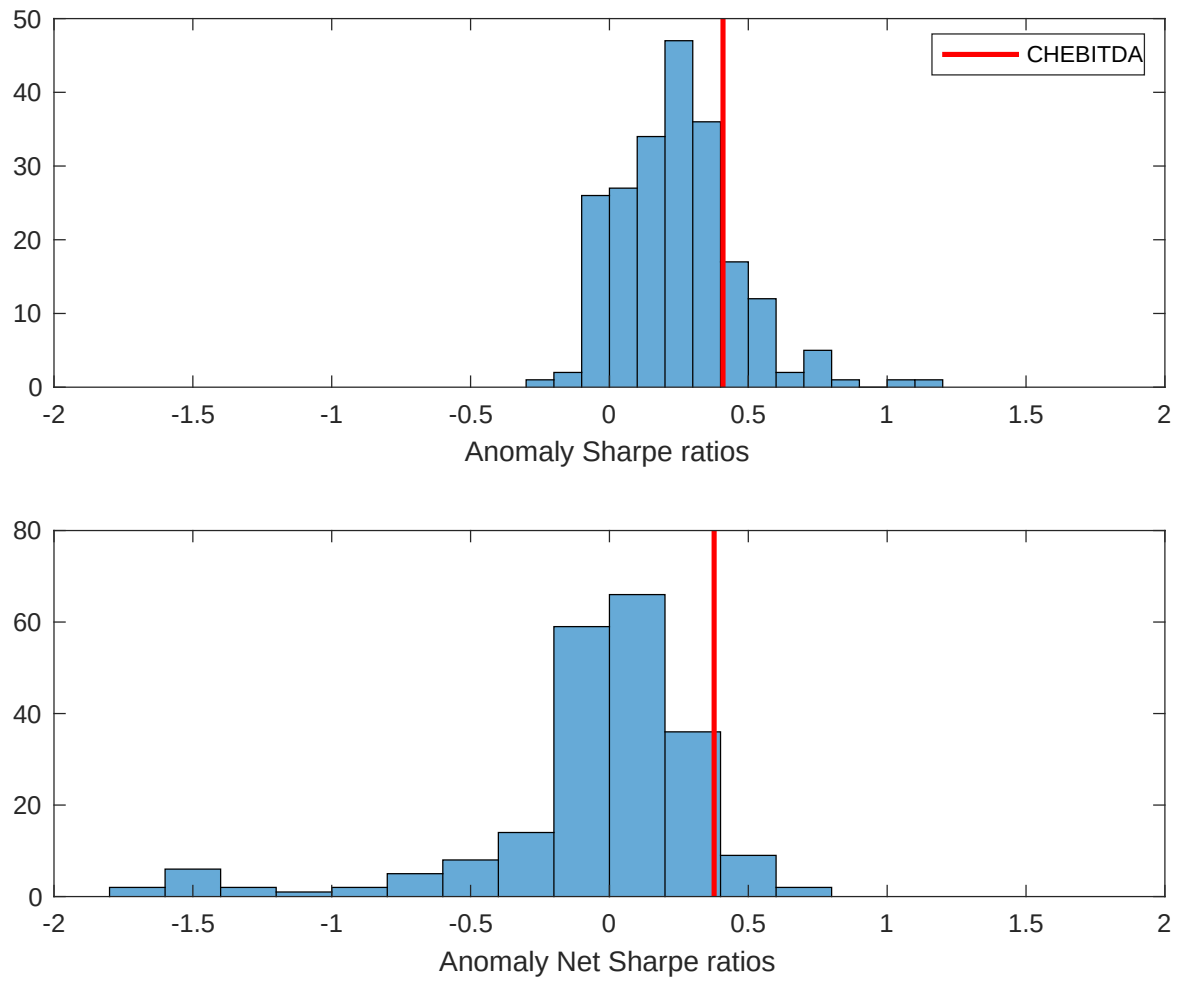


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CCC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

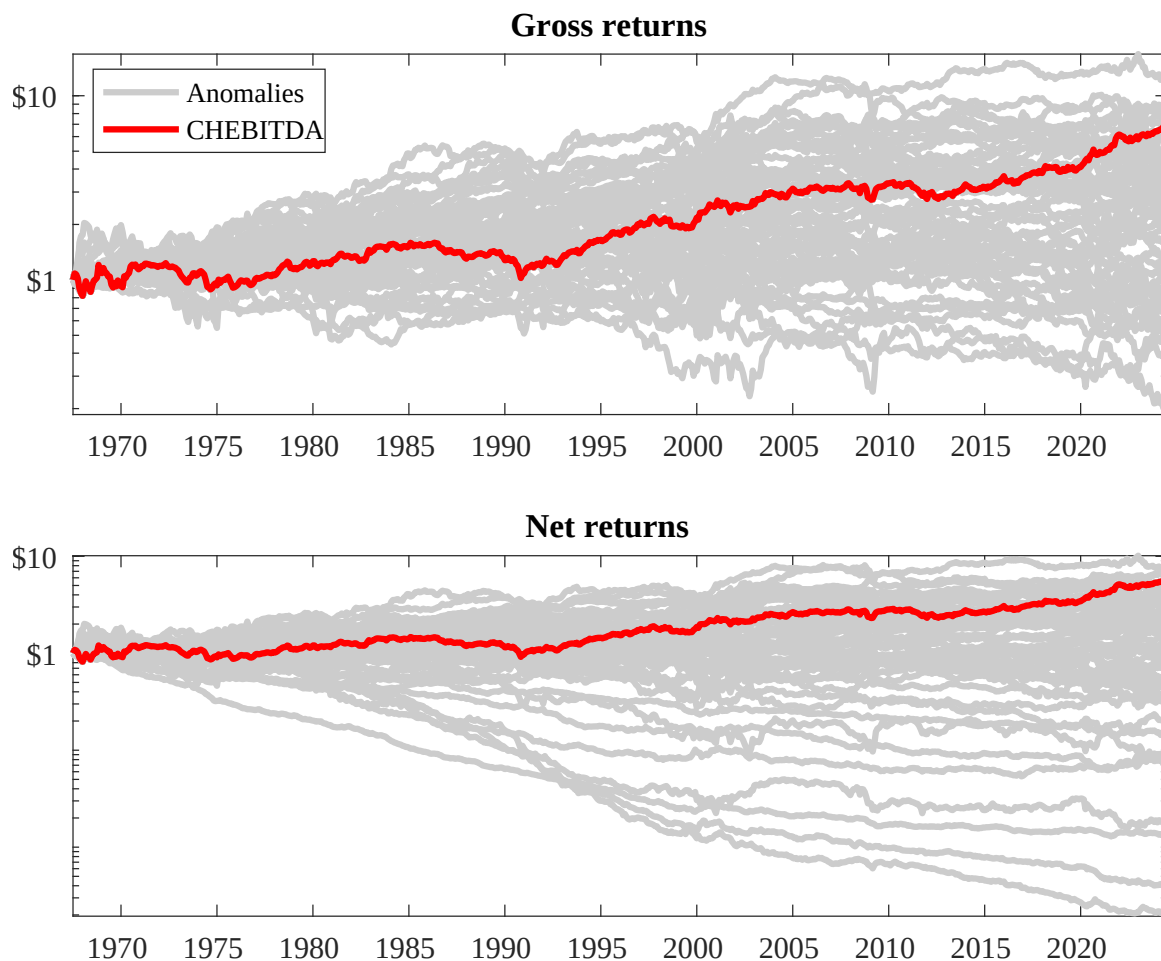
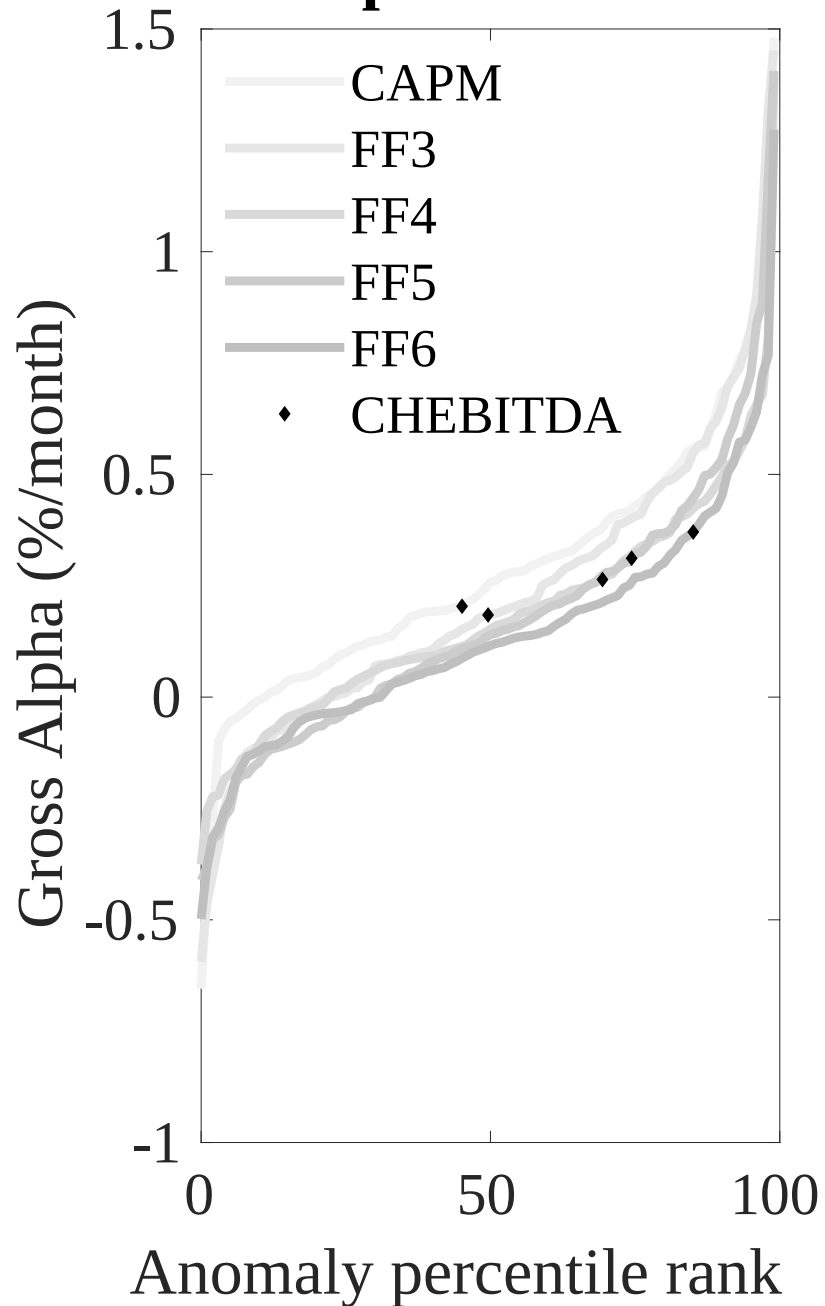


Figure 3: Dollar invested.

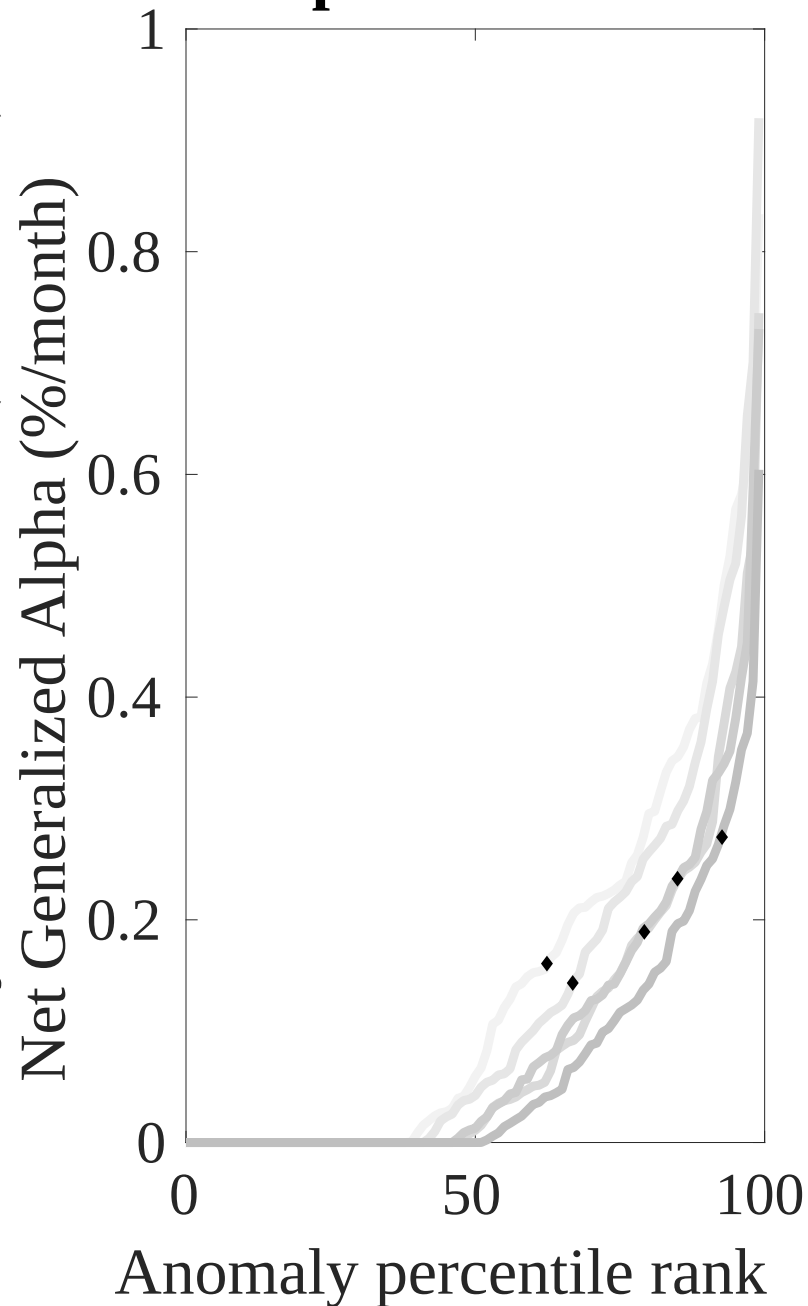
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CCC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



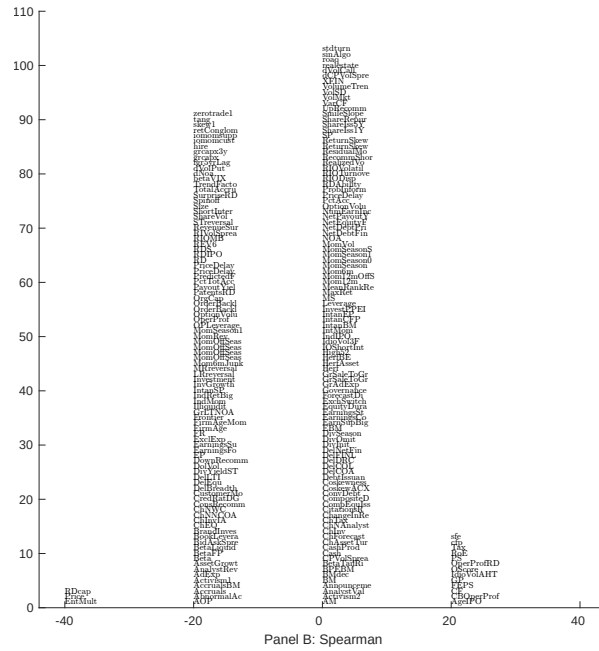
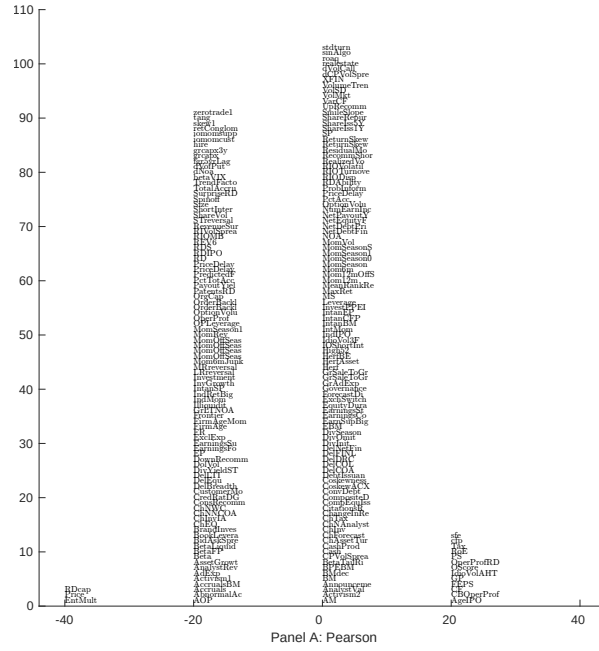


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CCC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

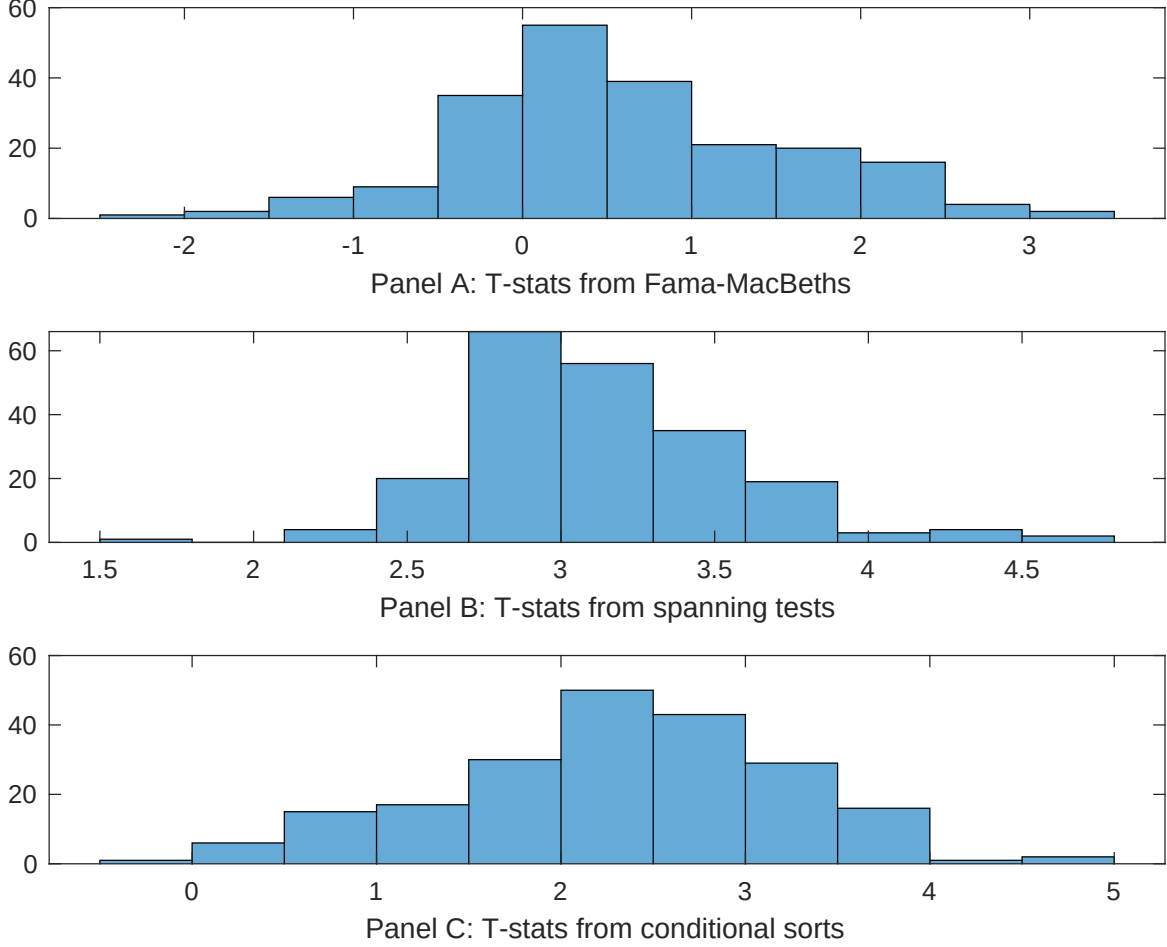


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CCC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.37]	0.14 [7.49]	0.12 [4.94]	0.13 [5.31]	0.15 [8.01]	0.12 [4.28]	0.16 [7.57]
CCC	0.58 [0.60]	-0.65 [-0.86]	-0.32 [-0.03]	0.22 [1.97]	-0.63 [-0.77]	0.14 [0.02]	0.55 [3.11]
Anomaly 1	0.84 [1.51]						0.12 [3.46]
Anomaly 2		0.95 [1.71]					-0.82 [-1.33]
Anomaly 3			0.47 [0.04]				0.10 [1.13]
Anomaly 4				0.19 [7.13]			0.12 [5.68]
Anomaly 5					0.16 [3.58]		0.27 [9.94]
Anomaly 6						0.61 [1.29]	0.17 [4.74]
# months	576	643	648	630	643	648	576
$\bar{R}^2(\%)$	2	2	0	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CCC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CCC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.37 [3.88]	0.34 [3.22]	0.36 [3.41]	0.33 [3.66]	0.34 [3.25]	0.36 [3.41]	0.37 [3.88]
Anomaly 1	4.75 [1.26]						7.58 [1.73]
Anomaly 2		1.01 [0.28]					-4.64 [-0.92]
Anomaly 3			-9.23 [-1.57]				-7.34 [-1.21]
Anomaly 4				2.46 [0.54]			9.37 [1.75]
Anomaly 5					-1.62 [-0.44]		-4.19 [-0.86]
Anomaly 6						-4.46 [-1.02]	-3.01 [-0.69]
mkt	18.46 [7.77]	18.85 [6.67]	17.01 [6.54]	17.51 [7.88]	17.90 [6.47]	19.05 [7.49]	16.52 [6.32]
smb	12.88 [3.04]	7.78 [1.53]	3.48 [0.86]	12.44 [3.56]	5.41 [1.13]	11.17 [1.90]	10.68 [1.95]
hml	6.73 [1.54]	5.52 [1.13]	4.79 [0.99]	8.46 [2.07]	6.28 [1.28]	6.42 [1.33]	9.47 [2.00]
rmw	-22.49 [-3.90]	-28.97 [-4.93]	-19.53 [-2.70]	-20.57 [-4.15]	-26.84 [-4.86]	-30.47 [-5.68]	-18.37 [-2.59]
cma	-26.19 [-4.09]	-7.73 [-1.06]	-9.61 [-1.33]	-26.61 [-3.91]	-6.71 [-0.92]	-7.35 [-1.03]	-31.13 [-4.13]
umd	-6.87 [-2.94]	-8.79 [-3.49]	-8.47 [-3.46]	-6.49 [-3.09]	-8.32 [-3.27]	-11.25 [-3.22]	-7.61 [-2.37]
# months	577	680	684	630	680	684	577
$\bar{R}^2(\%)$	27	22	22	29	22	22	28

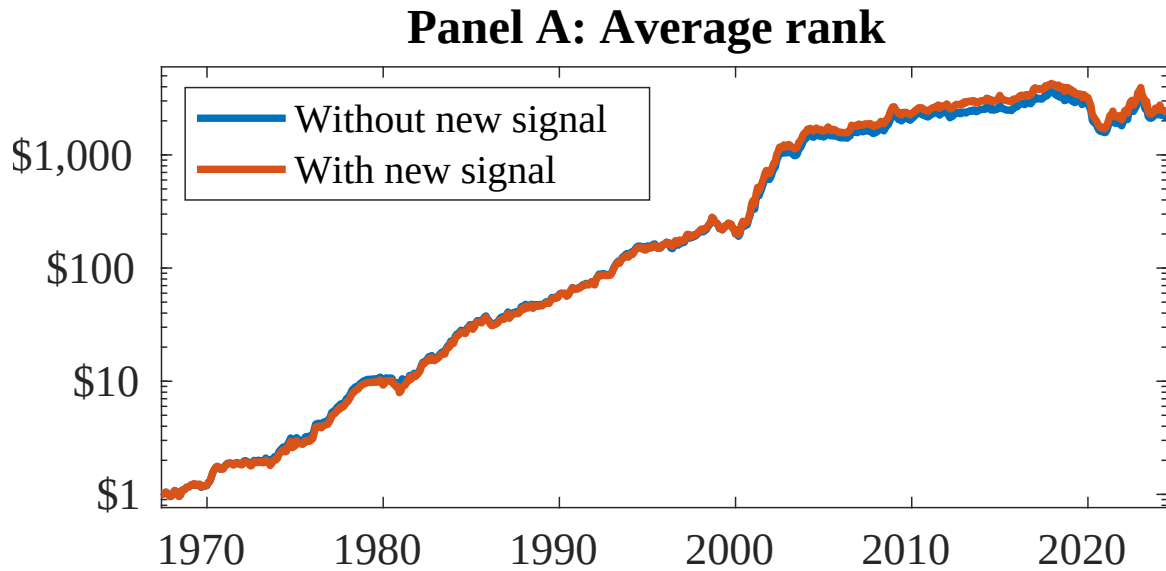


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CCC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Coval, J. D. and Moskowitz, T. J. (2001). The geography of investment: Informed trading and asset prices. *Journal of Political Economy*, 109(4):811–841.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.

- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Hou, K., Peng, L., and Xiong, W. (2009). A tale of two anomalies: The implications of investor attention for price and earnings momentum. *Working Paper*.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.
- Sloan, R. G. (1996). Do investors fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.